drawing the line

prof. dr. nicolas meseth

1 a world without steps
2 two cuts and a price
3 drawing your own lines

part 1

a world without steps

is this digital?



image: ai-generated (gpt-image-2)

digital is not the same as electronic

digital



flip scoreboard


light switch


abacus


a die

analog


slide rule


dial thermometer


dimmer


hourglass

not one of them contains a chip, so digital cannot mean electronic.

images: ai-generated (gpt-image-2)

digital means finitely many states, and nothing in between counts.

the second half carries the work, because it is what swallows disturbances.

the part no program controls

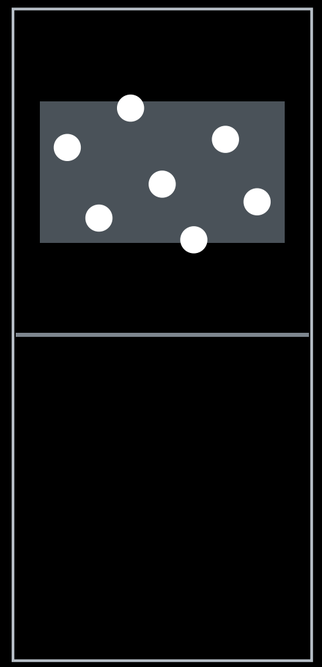


light is analog:
between any two brightnesses lies another.

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between any two brightnesses lies another.

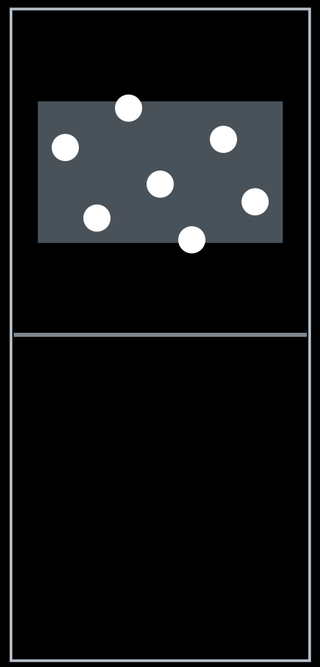
your program needs states:
finitely many, clearly apart.

same wobble, more regions

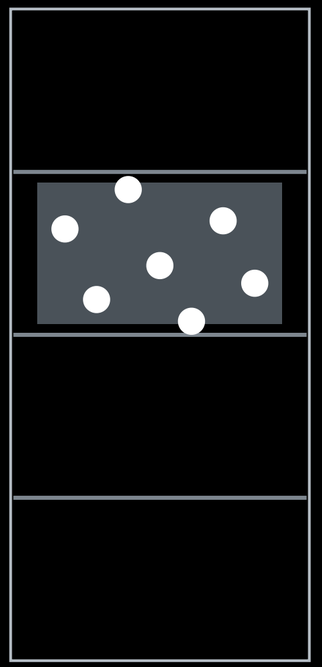


2 regions

same wobble, more regions

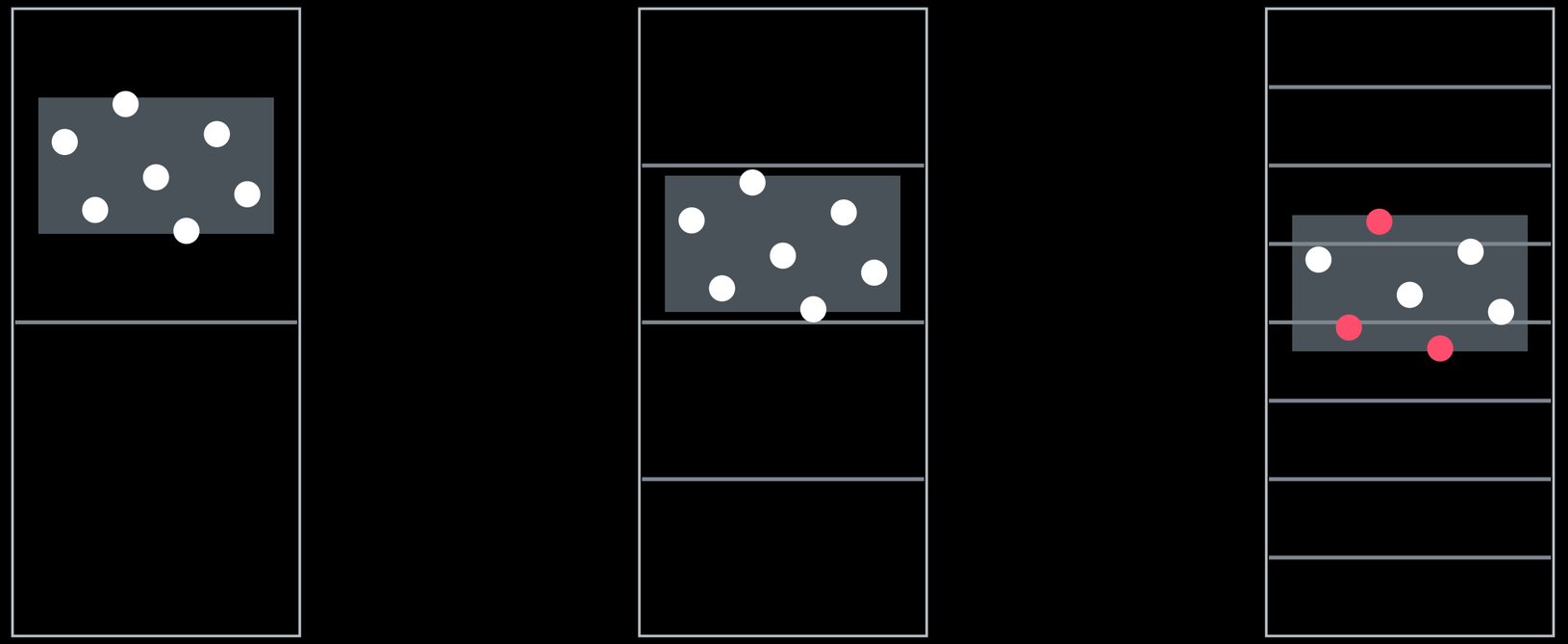


2 regions



4 regions

same wobble, more regions



2 regions

4 regions

8 regions

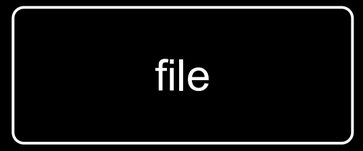
the wobble stays the same, but more regions leave each one a smaller margin.

what the thrown-away precision buys

analog: every copy carries the noise of the last one



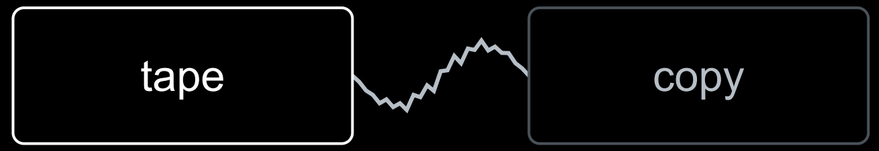
digital: every copy is read and set again



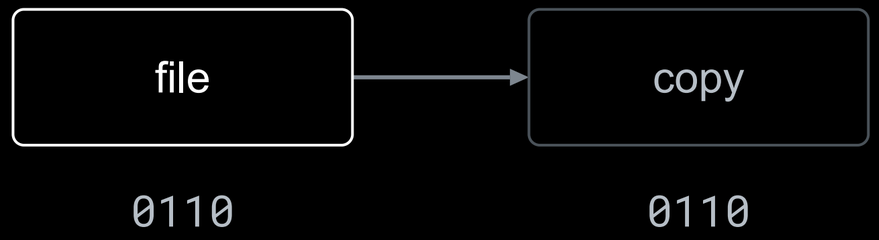

```
0110
```

what the thrown-away precision buys

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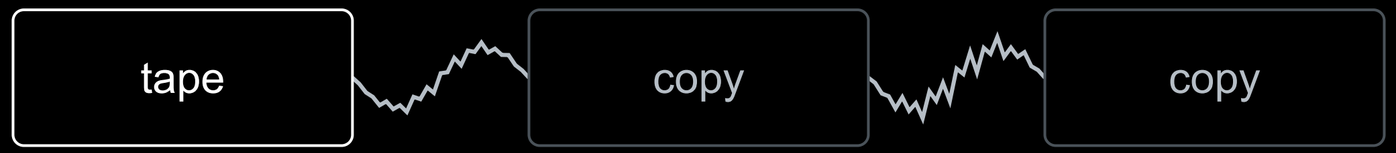


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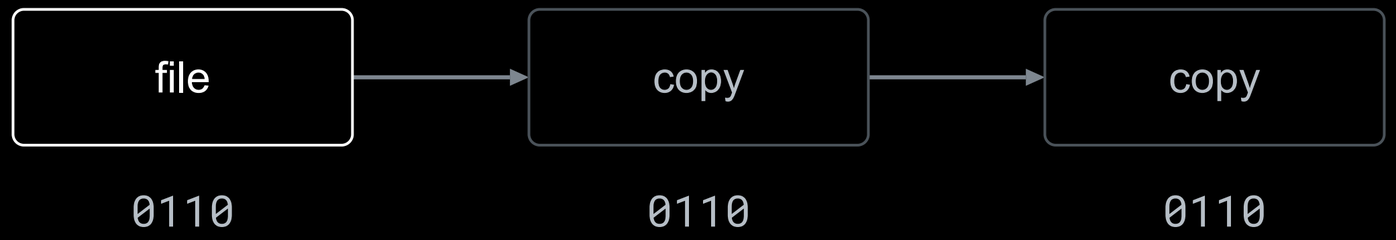


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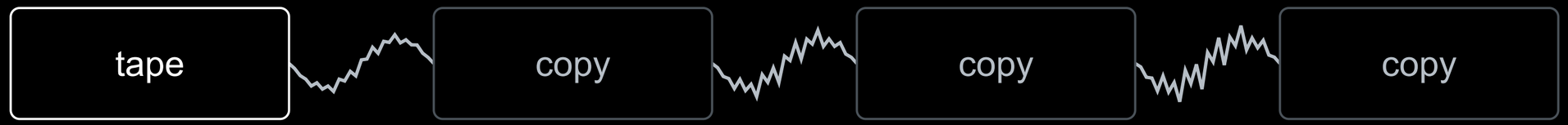


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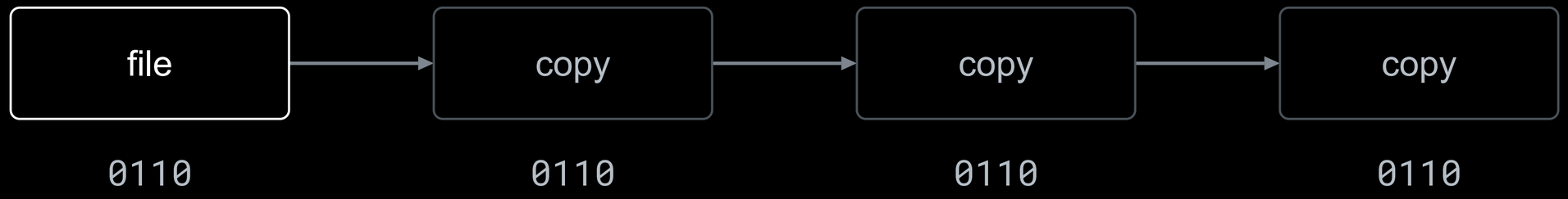


what the thrown-away precision buys

analog: every copy carries the noise of the last one



digital: every copy is read and set again



every digital copy is reborn from clean states, so it never inherits the noise.

part 2

two cuts and a price

digitising throws information away, and it does so on purpose.

which exact value inside a region it was is gone for good.

digitising takes two cuts

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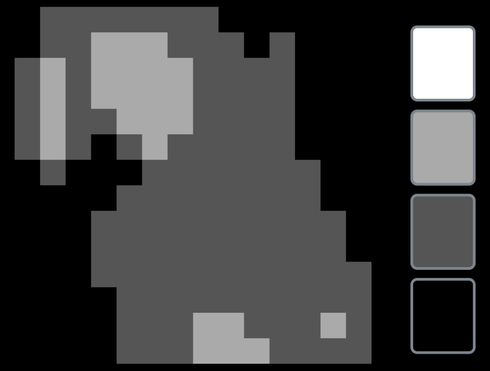
the analog picture



16 × 16 points



4 grey levels



cut 1: sampling
how many points: the resolution

cut 2: quantization
how many levels: the colour depth

sampling cuts the picture into points, and quantization cuts each point's value into levels.

which cut was too coarse?



too few points

blocks



too few levels

hard patches instead of soft shading

```
64 × 64 points × bits per point = size
```

points	bits per point	size	over the light link at 30 bit/s
4,096	24 bit	12 KB	about 55 minutes
4,096	1 bit	512 bytes	about 2 minutes

what you keep, you pay for in minutes.

how far can you go?

how far can you go?

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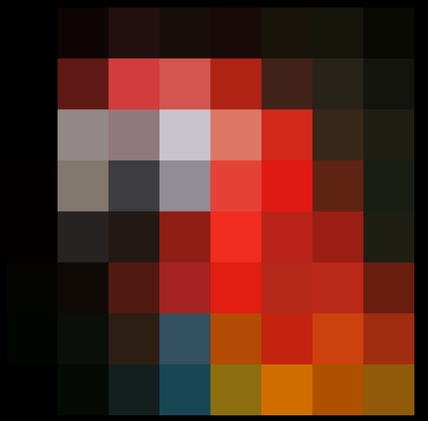
how far can you go?



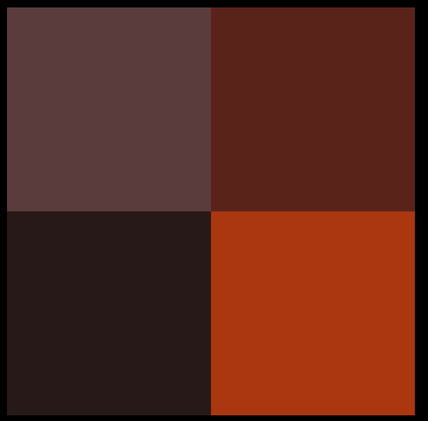
128 × 128
49,152 bytes



32 × 32
3,072 bytes



8 × 8
192 bytes



2 × 2
12 bytes

where would you stop?

the same two cuts, for sound

sampling slices time,
and quantization slices loudness.

a CD stores 44,100 samples per second with 16 bits each.

part 3

drawing your own lines

```
from lifi_hardware import sensor
r, g, b, c = sensor.read()

if c > 250:
    symbol = "bright"
else:
    symbol = "dark"
print(symbol)
```

this one line turns thousands of possible readings into just two symbols.

how many colours are safe?

two colours are safe, and eight are fast.

how many colours are safe?

two colours are safe,
and eight are fast.

somewhere in between sits your alphabet, and only your own measurements know where.

you decide where the lines go.

you decide where the lines go.

from there on, your decision is the truth, not the physics.